h-adic topology and profinite linear lambda terms

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1 Profinite Lambda terms and related concepts

One starts with the observation that if we have a work over a finite alphabet Σ:

$$W = a_{w_1} \dots a_{w_n} \in \Sigma^*$$

then one can represent W as a well-typed term in the simply typed lambda calculus:

$$a_{w_1} : q \to q, \dots, a_{w_n} : q \to q \vdash W : q \to q$$

where q is a base type that we informally interpret as a type of states, so we have a closed lambda term

$$\lambda a_{w_1}^{q \to q} \dots \lambda a_{w_n}^{q \to q} \lambda x^q . a_{w_k}(\dots (a_{w_1}\ q) \dots) : (q \to q) \dots (q \to q) \to q \to q$$

encoding a word w via the instance of the Church encoding. Further, we will call the type of such terms Church_Σ. Further, we have the monoid isomorphism:

$$\Sigma^* \cong \{M \text{ is a lambda term } | M : \text{Church}_\Sigma \}/\beta\eta$$

In other words, words over a finite alphabet can be viewed as lambda terms of a particular type factorised module the $\beta\eta$-equivalence. Generally, there is amount literature studying both syntactic and semantic aspects of the Church encoding of the finite words.

The claim made in [vGMM23] connects the finitary interpretation of STLC in the category of finite sets **FinSet** with deterministic finite automata as follows. Take any finite set Q, so for any type A, we define a set $[\![A]\!]_Q$ inductively defined as

- $[\![p]\!]_Q = Q$
- $[\![A \to B]\!]_Q = [\![A]\!]_Q \Rightarrow [\![B]\!]_Q$

so every term $x_1 : A_1, \ldots, x_n : A_n \vdash M : A$ transports to the function between finite sets $[\![M]\!] : [\![A_1]\!]_Q \times \ldots \times [\![A_n]\!]_Q \to [\![A]\!]_Q$. On closed terms, the interpretation induces a function $[\![.]\!]_Q : \Lambda_{\beta\eta}(A) \to [\![A]\!]_Q$ mapping every closed term M of type A to its interpretation $[\![M]\!]_Q$. Further, a word $W = a_{w_1} \ldots a_{w_k} \in \Sigma^*$ is mapped to the following morphism in **FinSet**: $[\![W]\!]_Q : (Q \Rightarrow Q) \times \ldots \times (Q \Rightarrow Q) \to (Q \Rightarrow Q)$, i.e.,

$$[\![W]\!]_Q : f_{w_1}, \ldots, f_{w_k} \mapsto f_{w_k} \circ \ldots \circ f_{w_1}$$

Recall that a *deterministic finite state automaton* on a finite alphabet $\Sigma = \{a_1, \ldots, a_n\}$ is a structure $\mathcal{A} = (Q, \delta, q_0, Acc)$, where Q is a finite set of states, $\delta : \Sigma \times Q \to Q$ is a transition function, $q_0 \in Q$ is the initial state and $Acc \subseteq Q$ is the set of accepting states. The transition function δ induces a family of functions $\delta_{a_i}(q_k) = \delta(a_i, q_k) : Q \to Q$ for each $a_i \in \Sigma$ and $q_k \in Q$. $\delta_{a_i}(q_k) = q_{k'}$ means that $\mathcal{A}$ transitions from q_k to q'_k encountering a_i. Now one can apply the interpretation $[\![W]\!]_Q$ to the functions $\delta_{a_{w_i}}$ for $1 \leq i \leq k$, so we obtain

$$\delta_w : [\![W]\!]_Q(\delta_{a_{w_1}}, \ldots, \delta_{a_{w_k}}) : Q \to Q$$

Therefore, given a deterministic finite automaton $\mathcal{A} = (Q, \delta, q_0, Acc)$, the tuple (Q, δ, q_0) induces the evaluation function

$$\mathrm{eval}_{\delta, q_0} : [\![\mathrm{Church}_\Sigma]\!]_Q \to Q$$

transporting every functional $F \in [\![\mathrm{Church}_\Sigma]\!]_Q$ to the state $F(\delta_{a_{w_1}}, \ldots, \delta_{a_{w_k}})$ in Q, therefore we have

$$\Sigma^* \cong \Lambda_{\beta\eta}(\mathrm{Church}_\Sigma) \to [\![\mathrm{Church}_\Sigma]\!]_Q \xrightarrow{\mathrm{eval}_{\delta, q_0}} Q$$

so we associate every finite word $w \in \Sigma^*$ with the final state $q_f = \delta_w(q_0)$. The inverse image $Acc \subseteq Q$ under this composite is called a *regular language* $L_\mathcal{A}$, the set of finite words recognisable by $\mathcal{A}$.

Regular languages $L_\mathcal{A}$ form a Boolean algebra $\mathrm{Reg}_Q(\mathrm{Church}_\Sigma)$ of regular languages of lambda terms of type Church_Σ recognised by Q. The algebra can be defined as the image of the Boolean algebra homomorphism $[\![.]\!]_Q^{-1} : 2^{[\![\mathrm{Church}_\Sigma]\!]} \to 2^{\Sigma^*}$ obtained by applying the contravariant powerset functor $2^{(.)} : \mathbf{FinSet}^{op} \to \mathbf{BA}$ to the semantic bracket $[\![.]\!]_Q$. The Boolean algebra $\mathrm{Reg}_Q(A)$ of regular languages of lambda terms of any higher order type A recognisable by a finite set Q is defined as the image of the Boolean algebra homomorphism $[\![.]\!]_Q^{-1} : 2^{[\![A]\!]_Q} \to 2^{\Lambda_{\beta\eta}(A)}$. That is, a set L of lambda terms of type A is recognisable by the finite set Q exactly when it has the form $[\![.]\!]_Q^{-1}(\mathrm{Acc}) = \{M \in \Lambda_{\beta\eta}(A) | [\![M]\!]_Q \in \mathrm{Acc}\}$ for some choice $\mathrm{Acc} \subseteq [\![A]\!]_Q$ of accepting elements. Thus, the collection $\mathrm{Reg}(A) \subseteq 2^{\Lambda_{\beta\eta}(A)}$ of lambda terms of type A as follows

$$\mathrm{Reg}(A) = \bigcup \{\mathrm{Reg}_Q(A) \mid Q \in \mathbf{FinSet}\}.$$

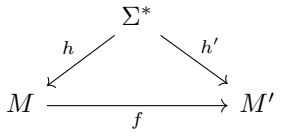
The monoid $\widehat{\Sigma^*}$ is the free profinite monoid generated by Σ. $\widehat{\Sigma^*}$ can be constructed as the limit, computed in the category of monoids **Mon**, of the projective system of finite monoid homomorphisms

$$(\Sigma^*/\phi \to \Sigma^*/\phi')_{\phi \subseteq \phi'}$$

where ϕ and ϕ' are finite index congruences on the free monoid Σ^* such that $\phi \subseteq \phi'$. Every finite index congruence ϕ, in turn, induces a surjective homomorphism $h : \Sigma^* \to M$ for the quotient monoid $M = \Sigma^*/\phi$. The surjectivity condition on h can be relaxed to show that $\widehat{\Sigma^*}$ of profinite words is the codirected limit of the composite functor

$$\Sigma^*/\mathbf{FinMon} \xrightarrow{\pi} \mathbf{FinMon} \hookrightarrow \mathbf{Mon}$$

$\Sigma^*/\mathbf{FinMon}$ is the slice category whose objects (M, h) are pairs consisting of a finite monoid M and a homomorphism $h : \Sigma^* \to M$ and whose morphisms $(M, h) \to (M', h')$ are homomorphisms $f : M \to M'$ such that the following triangle commutes:



The projection functor $\pi : \Sigma^*/\mathbf{FinMon} \to \mathbf{FinMon}$ transports (M, h) to its underlying monoid M. One obtains $\widehat{\Sigma^*}$ as the limit of a codirected diagram of monoid homomorphisms $(M \to M')_{(M,h)\to(M',h')}$ extending the diagram formed from the family $(\Sigma^*/\phi \to \Sigma^*/\phi')_{\phi \subseteq \phi'}$ to all homomorphisms $h : \Sigma^* \to M$ to a finite monoid M.

Recall that every homomorphism $h : \Sigma^* \to M$ induces a deterministic finite automaton with $Q := M$, $\delta(a, q) = q \cdot_M a$ for every $q \in M$ and $a \in \Sigma$. So every homomorphism $h : \Sigma^* \to M$ (assume $M = \{q_1, \ldots, q_m\}$) induces a decomposition of Σ^* into m components $L_{q_i} = h^{-1}(q_i)$ for $1 \leq i \leq m$ where each L_{q_i} is a regular language.

2 h-adic topology

The following summary is based on [Kas12, Chapter XVI].

Consider the complex algebra $K = \mathbb{C}[[h]]$ of complex formal series in one variable h. Every element $f \in \mathcal{K}$ has the form

$$f = \sum_{n=0}^{\infty} a_n h^n$$

where $a_n \in \mathbb{C}$ for each $n \in \mathbb{N}$. Every polynomial $f \in \mathbb{C}[h]$ is clearly a formal series. K form a commutative ring with operations given as

- $f + g = \sum_{n=0}^{\infty} a_n h^n + \sum_{n=0}^{\infty} b_n h^n = \sum_{n=0}^{\infty} (a_n + b_n) h^n$
- $f \cdot g = (\sum_{n=0}^{\infty} a_n h^n) \cdot (\sum_{n=0}^{\infty} b_n h^n) = \sum_{n=0}^{\infty} (\sum_{n=p+q} a_p b_q) h^n$

A formal series f is invertible iff its every coefficient is non-zero.

For any $n > 0$, consider the algebra $K_n = \mathbb{C}[h]/(h^n)$ of truncated polynomials obtained as the quotient of the ring of complex polynomials generated by the ideal (h^n). There is a morphism

$$\pi_n : K \to K_n$$

sending each formal series $\sum_{n=0}^{\infty} a_n h^n$ to the class of $\sum_{k=0}^{n-1} a_k h^k$ modulo (h^n). π_n is onto and its kernel is the ideal $h^n K$ generated by h^n in the ring of formal series. Therefore, we have the following isomorphism:

$$\mathbb{C}[[h]]/(h^n) \cong \mathbb{C}[h]/(h^n)$$

For $n > 0$, there is a surjective map of algebras $p_n : K_n \to K_{n-1}$ generated by the inclusion of ideals $(h^n) \subset (h^{n-1})$. So we have the inverse system of algebras $(K_n, p_n)_{n>0}$ and its inverse limit $\varprojlim_n K_n$. One has $p_n \circ \pi_n = \pi_{n-1}$ for each n. Also, there is a unique morphism of algebras from K to $\varprojlim_n K_n$ whose composition with the projection of the inverse limit is equal to π_n. Besides, the map $\pi : \mathbb{C}[[h]] \twoheadrightarrow \varprojlim_n \mathbb{C}[[h]]/(h^n)$ is the isomorphism, which is partly the instance of Birkhoff's subdirect representation theorem, [SB81, Theorem 8.6].

Let us equip K with the *h-adic topology*. The family of of subsets $\pi_n^{-1}(U_n)$ for $n > 0$ and $U_n \subseteq K_n$ form an open basis on K. $\{0\}$ is a family of open neighbourhoods in the discrete set K_n, the family $\pi_n^{-1}(0)$ is a family of open neighbourhoods of 0 in K for the h-adic topology. There is also the structure of a metric space on K given as follows. Given a formal series $f = \sum_{n=0}^{\infty} a_n h^n$, let $o(f)$ denote the unique natural number such that $a_{o(f)} \neq 0$ and $a_k = 0$ for each $k < o(f)$. For $f = 0$, set $o(f) = +\infty$ and extend $(\mathbb{N}, <)$ by adding the top element $+\infty$. It is easy to see that:

$$(h^n) = \{f \in K \mid o(f) > n - 1\}$$

and therefore

$$\bigcap_{n>0} (h^n) = \{0\}$$

We also have for any series $f, g \in K$:

$$o(f + g) \geq \min(o(f), o(g))$$

Now define $|.| : K \to \mathbb{R}_{\geq 0}$ as

$$|f| = 2^{-o(f)}$$

Then if we put $d(f,g) = |f-g|$, then $(\mathbb{C}[[h]], d)$ becomes an ultrametric space.

Let M be a left K-module, consider the family $(h^n M)_{n>0}$ of submodules of M with the canonical K-linear projections:

$$p_n : M_n = M/h^n M \to M_{n-1} = M/h^{n-1}M$$

They form an inverse system of K-modules, so we can consider the inverse limit

$$\widetilde{M} = \varprojlim_n M_n$$

having a natural structure of a left K-module. $\widetilde{M}$ has a natural topology as well, the inverse limit topology, for which the family of submodules $(h^n \widetilde{M})_n$ form an open basis. The projections $i_n : M \to M_n$ induce a unique K-linear map $i : M \to \widetilde{M}$ such that $\pi_n \circ i = i_n$ for each n. The kernel of i is given as

$$\mathrm{Ker}(i) = \bigcap_{n>0} h^n M$$

A K-module M is *separated* if $\mathrm{Ker}(i) = \{0\}$. It is *complete* if i is injective.

For any module K, the module $M/\mathrm{Ker}(i)$ is separated and the completion $\widetilde{M}$ of M is complete.

3 Term Assignment for Tensorial Logic

Figure 1: Term Calculus for Tensorial Logic

$$\frac{}{\vdash \mathbf{1} : \mathbb{1}}\ \mathbb{1}\mathbf{I} \qquad \frac{}{x : A \vdash x : A}\ \mathbf{ax} \qquad \frac{\Gamma \vdash M : \mathbb{1} \qquad \Delta \vdash N : B}{\Gamma, \Delta \vdash \mathbf{let}\ \mathbf{1} = M\ \mathbf{in}\ N : B}\ \mathbb{1}\mathbf{E}$$

$$\frac{x : A, \Gamma \vdash M : \bot}{\Gamma \vdash \nabla_l x.M : A \multimap \bot} \multimap \bot_{\mathbf{I}} \qquad \frac{\Delta \vdash N : A \qquad \Gamma \vdash M : A \multimap \bot}{\Gamma, \Delta \vdash N \triangleleft M : \bot} \multimap \mathbf{E}$$

$$\frac{\Gamma, x : A \vdash M : \bot}{\Gamma \vdash \nabla_r x.M : \bot \circ\!\!- A} \bot \circ\!\!-_{\mathbf{I}} \qquad \frac{\Gamma \vdash M : A \circ\!\!- \bot \qquad \Delta \vdash N : A}{\Gamma, \Delta \vdash M \triangleright N : \bot} \multimap \mathbf{E}$$

$$\frac{\Gamma \vdash M : A \qquad \Delta \vdash N : B}{\Gamma, \Delta \vdash M \otimes N : A \otimes B} \otimes\mathbf{I} \qquad \frac{\Gamma \vdash M : A \otimes B \qquad \Delta, x : A, y : B \vdash N : C}{\Delta, \Gamma \vdash \mathbf{let}\ x \otimes y\ =\ M\ \mathbf{in}\ N : C} \otimes\mathbf{E}$$

Definition 3.1. Reduction rules for tensorial logic:

$(\multimap \bot\beta)$ $N \triangleleft (\nabla_l x.M) \to M[x := N]$,

$(\bot \circ\!\!- \beta)$ $(\nabla_r x.M) \triangleright N \to M[x := N]$,

$(\mathbb{1}\beta)$ $\mathbf{let}\ \mathbf{1} = \mathbf{1}\ \mathbf{in}\ N \to N$,

$(\otimes\beta)$ $\mathbf{let}\ x \otimes y = M \otimes N\ \mathbf{in}\ P \to P[x := M, y := N]$,

$(\multimap \bot\eta)$ $(\nabla_l x.x \triangleleft M) \to M$ if x is not free in M,

$(\bot \circ\!\!- \eta)$ $(\nabla_r x.M \triangleright x) \to M$ if x is not free in M,

$(\otimes\eta)$ $\mathbf{let}\ x \otimes y = M\ \mathbf{in}\ N[z := x \otimes y] \to N[z := M]$,

$(\otimes_{\mathbf{nat}})$ $P[w := \mathbf{let}\ x \otimes y = M\ \mathbf{in}\ N] \to \mathbf{let}\ x \otimes y = M\ \mathbf{in}\ P[w := N]$,

$(\mathbb{1}_{\mathbf{nat}})$ $P[w := \mathbf{let}\ \mathbf{1} = M\ \mathbf{in}\ N] \to \mathbf{let}\ \mathbf{1} = M\ \mathbf{in}\ P[w := N]$.

4 On Dialogue Categories

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